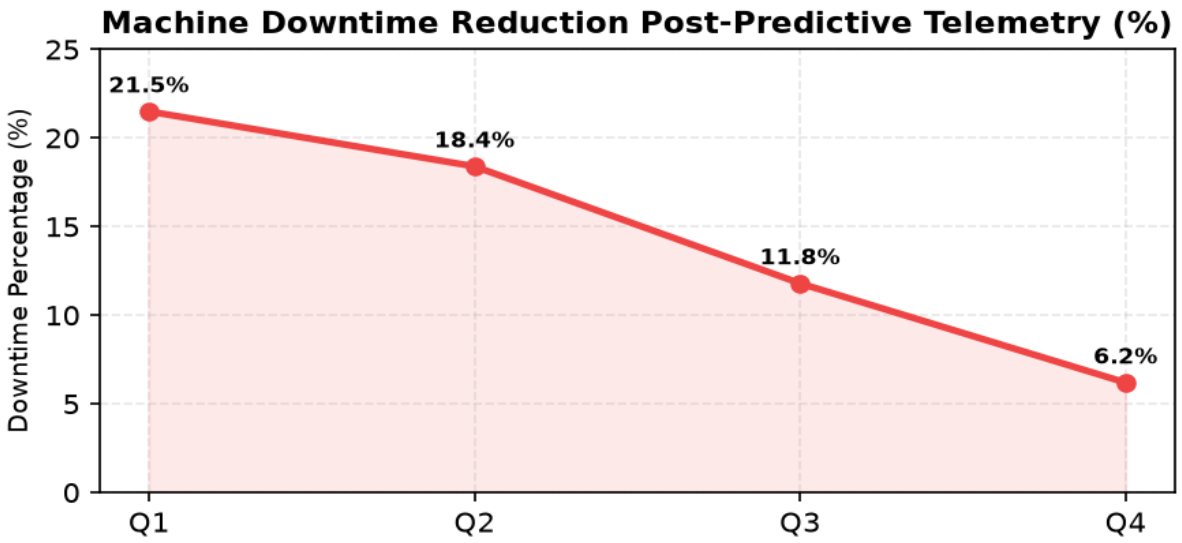
OPERATIONAL RELIABILITY & MAINTENANCE DIVISION

Plant Operations & Machine Downtime Report — FY2025

Section 1: Predictive Maintenance Overhaul. In late Q2, plant engineering initiated the deployment of IoT vibration sensors and predictive telemetry across 14 high-throughput stamping and assembly lines. Prior to this implementation, unexpected mechanical failures were the primary root cause of production bottlenecks.

Section 2: Machine Downtime Analysis by Quarter



Line Code	Equipment Type	Q2 Downtime (%)	Q4 Downtime (%)	Variance (pp)	Primary Corrective Action
L-101	Hydraulic Stamping Press	19.8%	5.4%	-14.4 pp	Vibration sensor predictive seal change
L-102	Robotic Welding Gantry	17.5%	6.1%	-11.4 pp	Automated servo calibration
L-103	Conveyor Transfer Cell	16.2%	6.8%	-9.4 pp	Belt tension telemetry upgrade
L-104	CNC Precision Mill	20.1%	6.5%	-13.6 pp	Spindle temperature telemetry
OVERALL	**Facility Average**	**18.4%**	**6.2%**	**-12.2 pp**	**Predictive maintenance system wide**

Proof Point: Unplanned machine downtime was reduced by 12.2 percentage points (from 18.4% in Q2 down to 6.2% in Q4), representing a 66.3% reduction in lost machine hours.

PLANT MAINTENANCE LOG — SCANNED ARCHIVE

[SCANNED DOCUMENT RECORD — INSPECTION SIGN-OFF] Date: 14-November-2025 | Shift: B | Auditor: R. Vance, PE Facility: Nexus Plant Alpha, Bay 4 EQUIPMENT MODIFICATION SUMMARY: Automated robotic transfer arms commissioned on Lines 101 through 104. Cycle time per unit pre-automation: 42.5 seconds (Q2 baseline). Cycle time post-automation: 32.3 seconds (Q4 verified). Cycle Time Reduction: -24.0% improvement in station flow rate. Physical inspection: ALL HYDRAULIC TOLERANCES WITHIN 0.02mm. Status: VERIFIED AND APPROVED FOR FULL PRODUCTION RUN.

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